



FEDERAL URDU UNIVERSITY OF
ARTS, SCIENCE & TECHNOLOGY

DEPARTMENT OF COMPUTER SCIENCE

SUBJECT: ARTIFICIAL
INTELLIGENCE

ASSIGNMENT: 01

DATED: 9-5-2026

SEAT NO:

23122093

STUDENT NAME:

MUHAMMAD ABEER ALI KHAN

FATHER NAME:

MUHAMMAD ARIF KHAN

SEMESTER (SECTION): 6TH (B)

MeetForge

AN INTELLIGENT MEETING MANAGEMENT & ACTION-PLANNING AGENT

1. Problem Identification

In modern organizational settings, meetings are the primary venue for strategy, decision-making, and task allocation. However, critical information is often discussed rapidly, leading to major operational challenges:

- **Task & Information Loss:** Key decisions, action items, assignees, and deadlines mentioned verbally are easily forgotten or overlooked after the meeting concludes.
- **Time Inefficiency:** Manually reviewing long meeting recordings, detailed transcripts, or informal notes consumes significant human effort and time.
- **Lack of Accountability:** Unclear assignment of responsibilities and untracked deadlines result in missed deliverables and project delays.

Solution: MeetForge solves these issues by acting as an intelligent meeting management agent that autonomously analyzes meeting discussions/transcripts, extracts structural data (topics, decisions, tasks, assignees, deadlines, and priorities), and compiles a coherent meeting summary and actionable follow-up plan.

2. Agent Formulation

A. Textual Formulation

MeetForge is formulated as a goal-based intelligent agent designed as follows:

Initial State: Raw, unstructured meeting input—such as audio/speech recordings, live transcripts, or text chat logs containing conversational meeting data.

Actions:

1. Receive and parse raw meeting conversation data or transcript.
2. Perform natural language processing and semantic analysis.
3. Identify core meeting topics and agenda items.
4. Extract concrete decisions made during discussions.
5. Detect explicit and implicit task items.
6. Map tasks to responsible individuals (assignees).
7. Extract associated deadlines, target completion dates, and set task priority levels.
8. Identify unresolved issues, risks, or open questions requiring follow-up.
9. Synthesize and format a structured meeting summary.
10. Generate action plans, push calendar invites, and set up automated follow-up reminders.

Goal State: A fully structured, accurate meeting report and action plan containing categorized topics, finalized decisions, assigned task items with assigned owners, deadlines, priorities, unresolved issues, and automated reminders.

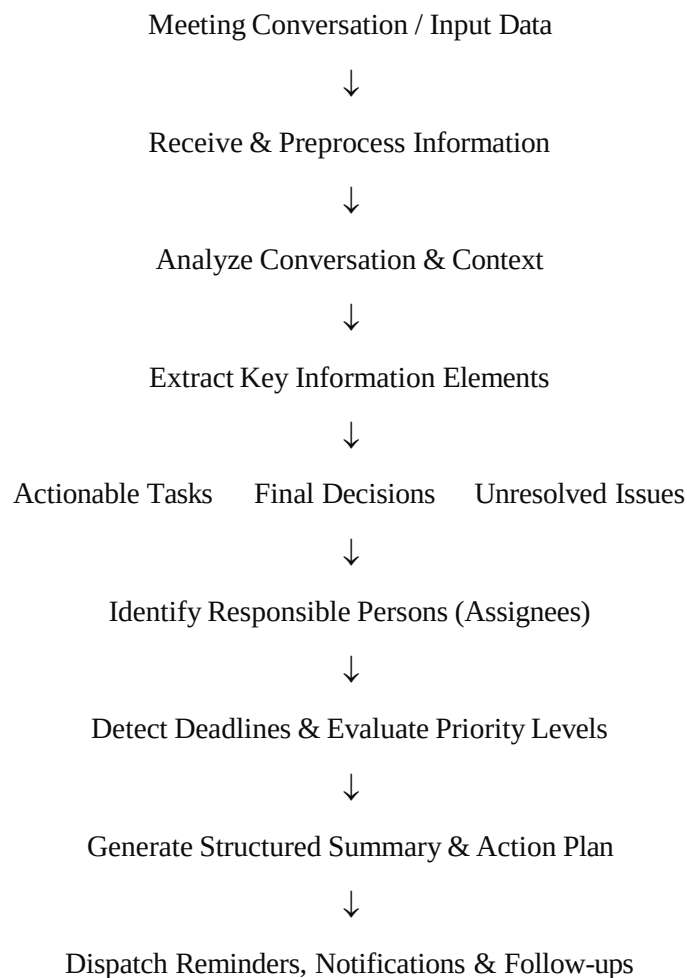
Performance Cost: The operational cost is evaluated based on processing latency, compute resource utilization, and error penalties (e.g., missed tasks, incorrect task assignments, wrong deadline extractions, inaccurate summary summaries, or inappropriate notification spam). The goal is to minimize these penalties.

Environment: Meetings, participants, conversational dynamics, text transcripts, calendars, organizational hierarchy, and historical meeting knowledge bases.

Sensors: Audio capture modules, speech-to-text engines, text inputs, chat logs, user profile directories, calendar integration APIs, and past meeting archives.

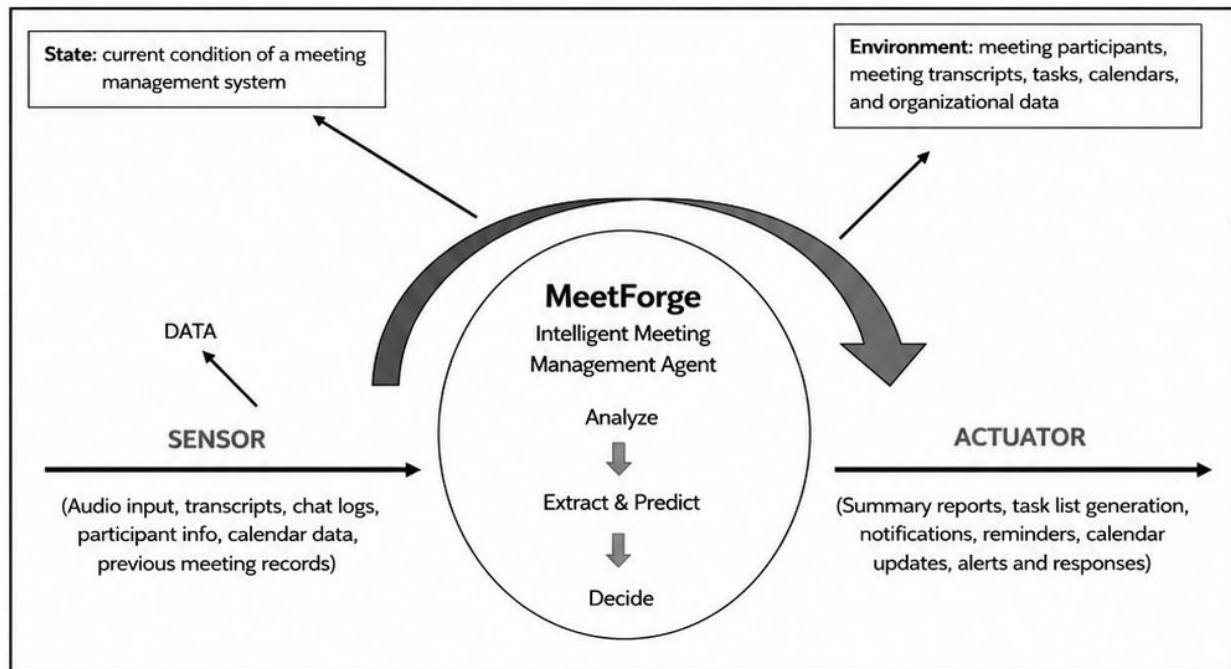
Actuators: Document/report generator, task management board integrators, notification/email dispatchers, calendar event updates, and follow-up reminder triggers.

B. Graphical Formulation



Graphical Representation of the MeetForge Agent:

3. GRAPHICAL PRESENTATION



3. Evaluation in the Context of PEAS Framework

PEAS stands for Performance Measure, Environment, Actuators, and Sensors.

PEAS Component	Description & Specification for MeetForge
Performance Measure	• High accuracy in extracting decisions, tasks, assignees, and deadlines. • Low rate of false positives (unnecessary tasks) and false negatives (missed tasks). • Conciseness and clarity of the generated meeting summary. • Timeliness of summary distribution and follow-up reminders. • User satisfaction score and low manual correction rate by participants.
Environment	• Meeting participants, speakers, and team structures. • Live spoken audio, meeting transcripts, and chat logs. • Task management platforms (e.g., Jira, Trello, Asana). • Calendar platforms (Google Calendar, Outlook) and historical meeting logs.
Actuators	• Summary and report output generation (PDF, Word, Dashboard). • Task Creation & Assignment System (auto-populating task lists). • Email & Push Notification dispatches for action items. • Calendar API integrations (creating follow-up events/deadlines). • Automated reminder and escalation scheduler.
Sensors	• Audio capture / Microphone stream feeds. • Automatic Speech Recognition (ASR) / Speech-to-Text processors. • Text input parsers (transcripts, chat logs, meeting notes). • User directory APIs (to resolve participant names/roles). • Calendar and previous meeting history data feeds.

4. Identification of Environmental Properties

Property	Classification	Reasoning for MeetForge Agent
Observability	Partially Observable	The agent only observes spoken words or written transcripts; it lacks access to offline side-conversations, non-verbal cues, unstated assumptions, or external context.
Agents Count	Multi-Agent	Operates in an environment driven by multiple human speakers/participants who interact, negotiate, assign tasks, and make collective decisions.
Determinism	Stochastic	Natural language speech features noise, interruptions, ambiguous expressions, and varying human behavior; the output cannot be deterministically predicted from raw audio alone.
Episodic / Sequential	Sequential	Earlier discussions, context, and context references directly influence later task assignments and decisions made within the meeting.
Dynamism	Dynamic	The environment changes dynamically during execution (e.g., speakers change topics abruptly, deadlines are altered mid-conversation, or tasks are reassigned).
Continuity	Mostly Continuous	Audio input and time flow continuously during live meetings, though text processing on transcripts occurs in discrete linguistic chunks.
Knowledge	Partially Known	The agent knows general operational language models and system rules, but does not possess complete knowledge of participant background contexts or future business decisions.
Accessibility	Partially Inaccessible	Private discussions, external email threads, and off-record decisions remain inaccessible to the agent.